Arbejdsopgaver - Optiske fænomener

Fysik B

1. Den blå himmel og den røde solnedgang

Se denne video: https://www.youtube.com/watch?v=9mnlJdABRns&t=39s

Læs nedenstående respons fra chatGPT på spørgsmålet om den røde solnedgang.

Skriv en forklaring på dansk, på ca. 5 linjer, som du selv kan relatere til. Stil evt. en AI et opfølgende spørgsmål undervejs. Tegn gerne en skitse af situationen på en tavle og tag et billede af det.

why is the sunshine red

That's a great question — the short answer is that **sunshine appears red at sunrise and sunset** because of the way Earth's atmosphere scatters light.

Here's a more detailed explanation:

- **Sunlight is made up of all colors** (a mix of wavelengths), which looks white when they're all combined.
- **Earth's atmosphere scatters shorter wavelengths (blue and violet) much more strongly** than longer wavelengths (red and orange). This is called **Rayleigh scattering**.
- When the Sun is **low on the horizon** (at sunrise or sunset), its light passes through a **much greater thickness of the atmosphere**.
- That longer path means most of the blue and green light is scattered away before reaching your eyes.
- What's left is mostly the **longer red and orange wavelengths**, so we see the sun and the surrounding sky as red, orange, or pink.

💡 Essentially:

Sunshine looks red at sunrise or sunset because most of the blue light gets scattered away, leaving mostly red light reaching your eyes.

Let me know if you'd also like to see a diagram or the math behind the scattering — I'd be happy to help!

2. De blå bjerge

Bjerge eller høje bygninger der er langt væk, ser ud til at være blå.

Skriv en kort forklaring på fænomenet (på dansk). Tag fx udgangspunkt i disse websider:

https://sky-lights.org/2015/07/27/qa-why-distant-mountains-look-blue/

https://www.reddit.com/r/askscience/comments/3ec6et/why_do_mountains_appear_blue_in_the_distance/

3. Normal luftspejling

Forklar fænomenet på figuren med egne ord.

Du må gerne bruge noten "Lys og optiske fænomener".

Se også denne video:
https://www.youtube.com/watch?v=pMMJo2q5ADM